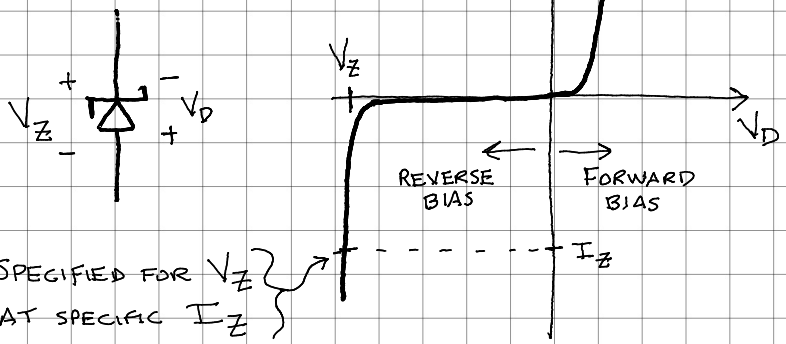
ZENER DIODES

W2AEW

(NOT ALL CREATED EQUAL)

BASICS



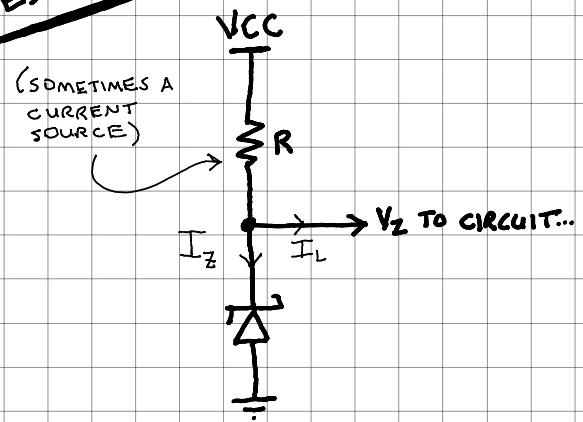
USES

- REVERSE BREAKDOWN IS USED FOR:
  - SHUNT REGULATOR
  - REFERENCE VOLTAGE
  - VOLTAGE CLAMP
  - ... MORE...

TYPICAL V_Z

2.4V TO 200V OR MORE

DESIGN INFO

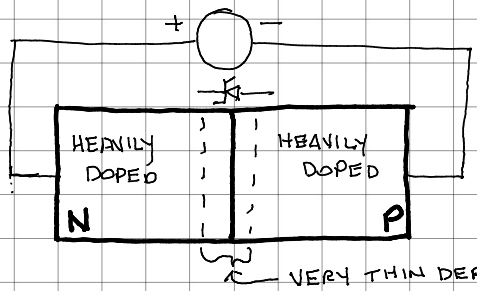


- DESIGN FOR SPECIFIC I_Z (ACCOUNT FOR LOAD)

$$R = \frac{(VCC - V_Z)}{(I_Z + I_L)}$$

ZENERS: NOT ALL CREATED EQUAL

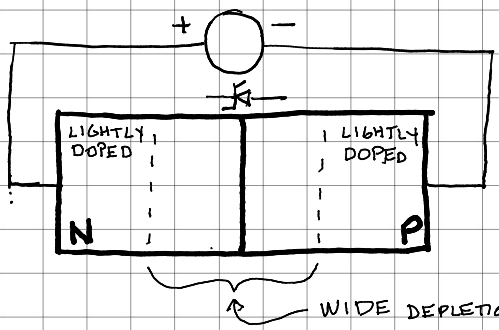
- ZENERS UP TO $V_Z \approx 5$ TO 6V: (TRUE ZENER DIODES)



ZENER

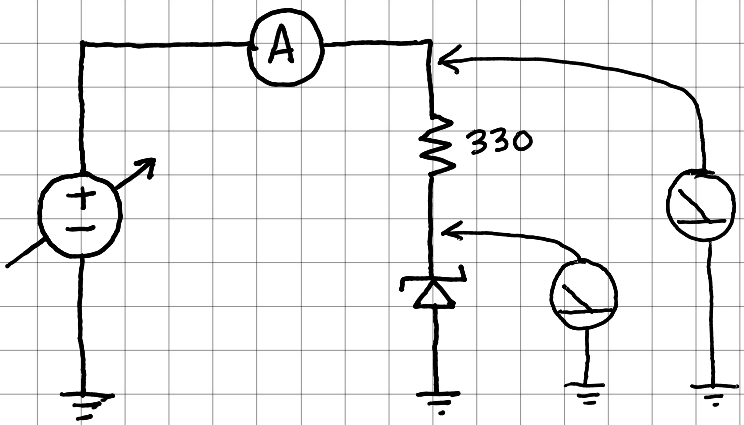
- ELECTRIC FIELD ACROSS DEPLETION REGION CAUSES CARRIERS TO "TUNNEL" THROUGH
- QUANTUM TUNNELING

- ZENERS WITH $V_Z > 5$ OR 6V: (AVALANCHE DIODES)



AVALANCHE

- ELECTRIC FIELD CAUSES TREMENDOUS ACCELERATION OF CARRIERS – FAST ENOUGH TO SMASH ELECTRONS FROM OTHER ATOMS, WHICH SMASH INTO MORE, AND SO ON...
- IMPACT IONIZATION

TEST CIRCUIT

W2AEW